

Unit 1

Historical Development

1. 1950s–1960s: Birth of Shared Computing

- Early computers were extremely large and expensive.
- Organizations used **mainframes** with **multiple terminals** that shared processing power.
- This concept of **time-sharing** allowed several users to access a single machine.
- It laid the foundation for the idea of delivering computing resources as a shared service.

Key Technologies:

- Mainframe computers
 - Time-sharing systems
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2. 1970s–1980s: Virtualization Concepts Emerge

- IBM introduced virtual machines (VMs), allowing one physical computer to run multiple operating systems.
- This led to better hardware utilization and isolation.

Key Events:

- 1972: IBM releases **VM/370**, enabling virtual machine technology.

- Virtualization becomes the core principle of modern cloud computing.
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3. 1990s: Rise of the Internet & Distributed Systems

- Wide adoption of the internet made remote access to computing resources possible.
- Concepts like **grid computing**, **distributed computing**, and **utility computing** emerged.
- Companies started thinking about computing as a metered service, similar to electricity.

Major Developments:

- 1997: The term “**cloud computing**” is believed to be used for the first time by Professor Ramnath Chellappa.
 - 1999: Salesforce introduced **software delivered over the internet (SaaS)**.
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4. 2000s: Modern Cloud Computing Takes Shape

a. Infrastructure-as-a-Service (IaaS)

- 2006: Amazon launched **AWS (Amazon Web Services)** with S3 and EC2.
- It became the first widely successful commercial cloud platform.
- Marked the start of on-demand computing infrastructure.

b. Platform-as-a-Service (PaaS)

- 2008: Google launched **Google App Engine**, enabling developers to deploy apps without managing servers.

c. Software-as-a-Service (SaaS) Expansion

- Companies like **Google**, **Microsoft**, and **Dropbox** popularized cloud-based applications.
 - Examples: Google Workspace, Office 365, Gmail.
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5. 2010s: Cloud Becomes Mainstream

- Cloud computing became essential for enterprises.
- Rise of **hybrid cloud**, **multi-cloud**, and **serverless computing**.
- Containers (e.g., Docker, Kubernetes) enabled scalable cloud-native applications.

Key Milestones:

- 2010: Microsoft Azure publicly launched.
 - 2013: Docker released—revolutionized container-based deployment.
 - 2014+: Kubernetes emerged as a standard for cloud orchestration.
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6. 2020s: Intelligent & Edge-Driven Cloud

- AI and cloud computing merged, enabling intelligent cloud services.
- Edge computing gained traction to reduce latency.

- Cloud providers expanded into ML and AI platforms (AWS SageMaker, Google Vertex AI, Azure AI).

Trends:

- Serverless architectures
- Edge + cloud hybrid models
- Cloud-native AI and automation
- Expansion of quantum cloud services